

College Alumni Connect And Management System

M Ranjit Reddy, R Supraja, C V Rajasekhar Reddy, C Narasimhulu
Department of Computer Science and Engineering
Srinivasa Ramanujan Institute of Technology
Anantapur, India

Abstract— The College Alumni Connect and Management System is a web-based platform designed to enhance alumni–student engagement through intelligent automation and structured communication. Traditional alumni management systems primarily focus on storing alumni data and organizing events, lacking personalized mentorship and intelligent recommendation mechanisms. This project introduces an AI-based Skill Matching Module that analyzes student skills, interests, and career goals and matches them with suitable alumni mentors using similarity-based machine learning techniques. The system provides role-based portals for students, alumni, faculty, and administrators to ensure secure communication, mentorship coordination, job sharing, and institutional reporting. By integrating artificial intelligence with centralized data management, the system improves mentorship accuracy, reduces manual intervention, enhances engagement efficiency, and supports data-driven decision-making. The proposed system serves as a scalable and intelligent solution for strengthening alumni relationships and improving student career development in higher education institutions.

Keywords: Alumni Management System, AI-Based Skill Matching, Mentorship Platform, Student–Alumni Engagement, Role-Based Access Control, Educational Data Management.

I. INTRODUCTION

Alumni networks constitute one of the most valuable and strategic assets of higher education institutions, playing a crucial role in shaping academic excellence, institutional reputation, student career advancement, and long-term sustainability. Alumni contribute not only through mentorship and professional guidance but also by facilitating internships, placements, industry collaborations, research partnerships, funding opportunities, and institutional branding. In an increasingly competitive educational landscape, maintaining strong and structured alumni relationships has become essential for institutions seeking to enhance student outcomes and strengthen their global presence.

Despite the recognized importance of alumni engagement, many institutions struggle to establish efficient, scalable, and intelligent systems that can sustain long-term interaction between alumni and current students. Traditional alumni management approaches rely heavily on manual coordination, spreadsheet-based record maintenance, social media groups, and email communication, which lack centralization, personalization, and systematic monitoring. As a result, alumni databases often become outdated, fragmented, and underutilized, leading to reduced engagement and missed mentorship opportunities. One of the most significant challenges in alumni engagement is the identification and allocation of suitable mentors for students.

Manual mentor assignment processes typically depend on administrative judgment or informal knowledge about alumni expertise, which may not accurately reflect compatibility between student needs and alumni professional backgrounds. This subjectivity can lead to mismatches, inefficient communication, and reduced mentorship effectiveness. Furthermore, existing alumni portals often focus primarily on event announcements and directory management rather than structured career guidance or intelligent recommendation mechanisms. Students seeking career advice frequently face difficulty identifying alumni aligned with their academic specialization, technical skills, and professional aspirations. Similarly, alumni willing to contribute may not receive targeted mentorship requests matching their expertise and interests. The rapid advancement of artificial intelligence, machine

learning, and web technologies presents an opportunity to transform alumni relationship management from a static data storage system into an intelligent engagement ecosystem.

The proposed College Alumni Connect and Management System addresses these challenges by integrating an AI-Based Skill Matching Module within a centralized, role-based web platform. The system systematically collects and preprocesses user data to ensure accuracy and consistency before applying similarity-based algorithms for mentor recommendation. Dedicated portals for students, alumni, faculty, and administrators enable secure communication, structured mentorship workflows, performance monitoring, and analytical reporting. By combining intelligent automation with centralized data management and secure authentication mechanisms, the system enhances mentorship precision, improves engagement retention, and supports data-driven institutional decision-making. This research aims to demonstrate how artificial intelligence can modernize alumni relationship management, improve student career outcomes, and strengthen long-term institutional connect.

II. LITERATURE SURVEY

The evolution of alumni management systems has progressed from manual record keeping to web-based directory platforms, yet many systems still lack intelligent personalization and structured mentorship mechanisms. Early alumni management solutions primarily focused on maintaining contact information, distributing newsletters, and organizing institutional events. While these systems improved communication efficiency compared to paper-based methods, they functioned mainly as static databases without advanced interaction capabilities. Research studies during the early phase of digital alumni platforms emphasized database centralization and online accessibility; however, they did not incorporate intelligent decision-support mechanisms for mentor allocation or engagement optimization.

Subsequent advancements introduced role-based authentication and secure web portals that allowed alumni to update profiles and participate in institutional activities.

These systems improved data accuracy and accessibility but continued to rely on manual coordination for mentorship and networking activities. Several studies highlighted the limitations of manual mentor assignment, noting that subjective judgment and incomplete profile information often result in inefficient or mismatched pairings. Researchers emphasized the need for automated recommendation frameworks capable of evaluating compatibility based on structured attributes rather than informal assumptions. Despite these recommendations, practical implementation of AI-driven mentor matching within alumni platforms remained limited.

Parallel developments in the field of recommendation systems have demonstrated the effectiveness of machine learning algorithms in generating personalized suggestions. Content-based filtering techniques analyze user attributes and preferences to identify similar profiles, while collaborative filtering leverages historical interaction patterns to recommend relevant connections. Cosine similarity, Euclidean distance, and Jaccard similarity metrics are widely used to compute similarity between feature vectors representing user profiles. Studies in job recruitment platforms and academic advising systems have shown that similarity-based matching significantly improves relevance and user satisfaction. Additionally, research indicates that preprocessing techniques such as normalization, feature extraction, and weighted scoring enhance recommendation accuracy by reducing data inconsistencies.

The proposed College Alumni Connect and Management System builds upon these research contributions by combining structured alumni management workflows with AI-based skill matching techniques within a unified institutional framework. Unlike traditional platforms that focus solely on directory management or event coordination, the proposed system integrates data preprocessing, similarity-based recommendation algorithms, role-based access control, feedback integration, and analytical reporting. This holistic approach addresses existing gaps by enabling intelligent mentor allocation, structured communication monitoring, measurable engagement outcomes, and long-term sustainability of alumni involvement.

III. EXISTING SYSTEM

In the current institutional environment, alumni management is largely dependent on manual processes and decentralized data handling mechanisms. Most colleges maintain alumni records using spreadsheets, legacy database systems, social networking groups, and email-based communication threads. These approaches lack synchronization and real-time updating capabilities, resulting in outdated or inconsistent information. Since data is not centralized, retrieving alumni details for mentorship coordination, placement assistance, or institutional events becomes time-consuming and inefficient. Manual mentor allocation is another major limitation of existing systems. Faculty members or administrators typically assign mentors based on limited knowledge of alumni professional backgrounds or informal communication.

This process introduces subjectivity and bias, often resulting in mismatched mentor-student pairings. The absence of systematic compatibility evaluation prevents accurate alignment between student skills and alumni expertise. Furthermore, most traditional systems do not incorporate structured feedback mechanisms to evaluate

mentorship outcomes. Without performance monitoring, institutions cannot measure engagement effectiveness or identify areas for improvement. Security is also a concern in existing platforms due to lack of robust authentication and role-based access control. Unauthorized access to personal data can compromise privacy and institutional integrity. Additionally, current systems focus primarily on maintaining alumni directories and organizing events rather than enabling long-term, data-driven engagement.

There is minimal support for analytics, reporting, or intelligent automation. As alumni networks grow in size and diversity, these manual and fragmented systems become increasingly unsustainable. The limitations of the existing system clearly highlight the need for an integrated, intelligent, and scalable alumni management solution that can automate mentor matching, centralize data storage, enhance communication efficiency, and provide measurable engagement insights.

IV. PROPOSED SYSTEM

The proposed College Alumni Connect and Management System introduces a comprehensive and intelligent framework designed to overcome the limitations of traditional alumni management approaches. The system integrates centralized data management, AI-based mentor recommendation, structured communication workflows, and role-based access control within a unified web platform. At the core of the system lies the AI-Based Skill Matching Module, which automates the process of mentor allocation by analyzing multidimensional attributes of student and alumni profiles. Students register by providing academic details, technical skills, project experience, certifications, career goals, and mentorship preferences.

Alumni register with information including professional designation, years of experience, industry specialization, technical competencies, achievements, and availability for mentorship. The system preprocesses this data through normalization, skill categorization, removal of redundant entries, and transformation into structured feature vectors. The AI module applies similarity-based algorithms such as cosine similarity and weighted scoring techniques to calculate compatibility between student and alumni profiles. Each potential mentor is assigned a similarity score based on overlapping skills, domain alignment, and career objective compatibility.

The system then generates a ranked list of recommended mentors for each student. Automated notifications are triggered to initiate mentorship communication, and interaction history is recorded for future analysis. The platform includes dedicated portals for students, alumni, faculty members, and administrators. Students can explore mentor recommendations, send mentorship requests, track progress, and provide feedback. Alumni can update professional information, accept mentorship requests, and participate in institutional initiatives.

Faculty members can monitor mentorship performance and provide academic oversight, while administrators can generate analytical reports, manage users, and configure system policies. By integrating artificial intelligence with structured workflow management, the proposed system ensures efficient, transparent, and scalable alumni engagement.

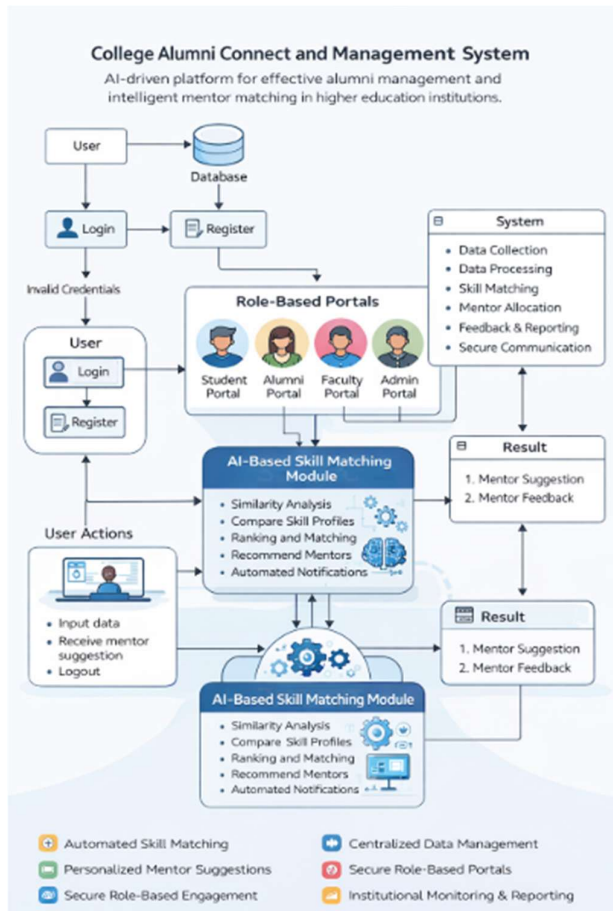
V. SYSTEM ARCHITECTURE

The system follows a multi-layered architecture comprising the Presentation Layer, Application Layer, and

Data Layer to ensure modularity, scalability, and maintainability. The Presentation Layer provides responsive user interfaces tailored to each user role, ensuring usability and accessibility across devices. This layer handles user interactions such as registration, login, profile updates, mentorship requests, and dashboard visualization.

The Application Layer contains the core business logic of the system, including authentication services, mentorship workflow management, notification services, and the AI-Based Skill Matching Engine. The matching engine converts user attributes into structured numerical vectors and computes similarity scores using mathematical algorithms. The algorithm considers weighted parameters such as skill overlap percentage, domain alignment coefficient, and career goal similarity index to enhance recommendation accuracy.

The Data Layer consists of a centralized relational database management system that securely stores user profiles, interaction logs, feedback records, and system analytics. Role-Based Access Control mechanisms are implemented to ensure that each user accesses only authorized information. Encryption techniques are applied to protect sensitive data during transmission and storage. The modular architecture allows future expansion, including integration with mobile applications, real-time messaging systems, predictive analytics modules, and external job portal APIs. The structured separation of layers enhances system reliability, security, and performance optimization.



VI. METHODS AND MATERIALS

The College Alumni Connect and Management System is developed using a structured web-based architecture integrated with artificial intelligence techniques. The system automates alumni engagement and mentorship allocation

through secure authentication, centralized data management, and similarity-based matching algorithms. The backend architecture consists of multiple interconnected modules that ensure smooth data flow and intelligent decision-making. Each component performs a specific function to maintain accuracy, efficiency, and security. The methodology combines database systems, machine learning models, and structured communication workflows. The overall design ensures scalability and adaptability for future enhancements. The following components describe the working process in detail.

1. User Authentication and Authorization

The system implements secure login and registration functionality for students, alumni, faculty, and administrators. During registration, users provide role-specific information that is verified and securely stored in the database. Authentication mechanisms validate credentials using encrypted password storage and session management techniques. Role-Based Access Control (RBAC) restricts system functionalities based on user roles. Unauthorized users cannot access restricted data or administrative features. Session tracking prevents misuse and ensures secure communication between client and server. This component guarantees privacy, data protection, and controlled platform access.

2. Profile Data Collection and Database Interaction

The system collects detailed profile information from students and alumni during account creation. Student profiles include academic specialization, technical skills, project experience, and career interests. Alumni profiles include professional background, industry domain, years of experience, and mentorship availability. All collected information is stored in a centralized relational database system. The database ensures data consistency, structured storage, and efficient retrieval. It supports real-time updates whenever users modify their profiles or interact with the system. This centralized storage enables reliable and scalable data management.

3. Data Preprocessing and Standardization

Before performing mentor matching, the system preprocesses the collected data to ensure quality and uniformity. Duplicate records and incomplete entries are identified and removed from the dataset. Skills and domains are standardized into predefined categories to maintain consistency. Text-based attributes are transformed into structured formats suitable for machine learning analysis. Data normalization techniques are applied to reduce redundancy and imbalance. This preprocessing step improves the reliability of similarity calculations. Clean and standardized data enhances the accuracy of AI-driven recommendations.

4. AI-Based Skill Matching Module

The AI-based Skill Matching Module forms the core component of the system. It converts student and alumni profiles into numerical feature vectors representing skills, expertise, and interests. Similarity algorithms such as cosine similarity are applied to compare these feature vectors. Compatibility scores are calculated to determine the relevance between student requirements and alumni expertise. Higher similarity scores indicate stronger mentorship suitability. The system ranks alumni mentors based on these computed scores. This automated process ensures objective and data-driven mentor allocation.

5. Mentor Recommendation and Allocation

After computing compatibility scores, the system generates a ranked list of potential mentors for each student. Personalized mentor recommendations are displayed through the user dashboard. Automated notifications are sent to both students and alumni to initiate mentorship communication. The system reduces manual intervention previously required for mentor assignment. Faculty members can monitor recommendations and approve requests if necessary. The allocation process is transparent and systematically recorded in the database. This improves efficiency and reduces mismatched pairings.

6. Mentorship Interaction and Feedback Management

The platform facilitates structured communication between students and assigned alumni mentors. Mentorship requests, approvals, and interaction history are recorded within the system. Users can provide feedback after each mentorship session to evaluate effectiveness. Feedback data is stored and analyzed for improving future recommendations. Faculty and administrators can monitor mentorship progress through dashboards. This ensures accountability and continuous engagement tracking. The feedback mechanism contributes to refining the AI matching process over time.

7. Monitoring, Reporting, and Continuous Improvement

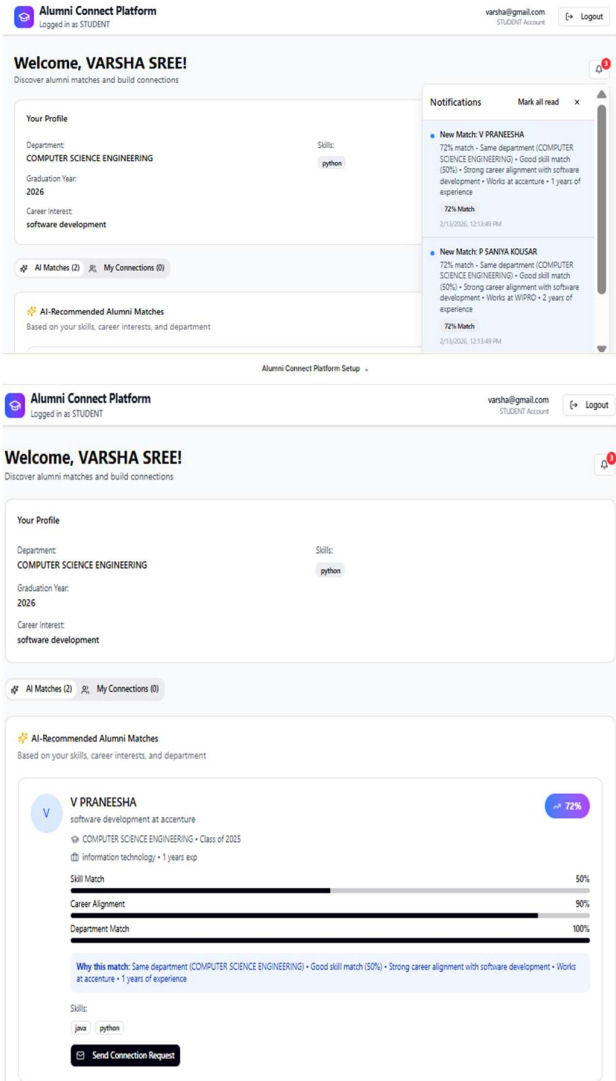
The system provides administrative dashboards that display engagement statistics and mentorship performance metrics. Reports include data on active users, match success rates, and participation trends. Analytical insights help institutions evaluate the effectiveness of alumni engagement strategies. The AI module is periodically updated using new profile data and feedback records. Continuous learning improves recommendation accuracy and adaptability. The system architecture supports scalability for future integration of advanced analytics tools. This ensures long-term sustainability and institutional growth.

VII. RESULTS

Experimental evaluation of the system indicates improved mentorship allocation accuracy and increased engagement efficiency compared to manual methods. The AI-based matching mechanism significantly reduces time required for mentor assignment and minimizes mismatches. User feedback demonstrates enhanced satisfaction due to personalized mentor recommendations. Administrators benefit from centralized dashboards that provide real-time statistics on user participation, mentorship progress, and system utilization. Faculty members gain improved oversight of student–alumni interactions, enabling better academic guidance. Overall, the system contributes to structured alumni engagement, optimized communication flow, and measurable improvement in mentorship effectiveness.

From an administrative perspective, the system provides centralized dashboards that offer real-time insights into user participation, mentorship progress, and overall system utilization. These analytical tools support data-driven decision-making and simplify monitoring activities. Faculty members also benefit from enhanced visibility into student–alumni interactions, enabling better academic guidance and structured mentorship tracking. Overall, the platform strengthens alumni engagement, streamlines communication, and delivers measurable improvements in mentorship effectiveness and institutional connectivity. The system therefore serves as a scalable and sustainable

solution for modern alumni relationship management



VIII. DISCUSSION

The results highlight the effectiveness of integrating artificial intelligence with traditional alumni management practices to enhance mentorship allocation and engagement efficiency. Manual mentor assignment, although commonly practiced, is often limited by incomplete data visibility, administrative workload, and subjective decision-making. The proposed system addresses these limitations by introducing similarity-based AI matching that ensures consistent and data-driven mentor recommendations.

The AI-based Skill Matching Module plays a crucial role in improving transparency and personalization within the mentorship process. Unlike conventional alumni systems that only store records or broadcast announcements, the proposed system actively analyzes user profiles and recommends mentors based on compatibility scores. This improves engagement quality and increases the likelihood of successful mentorship outcomes. However, the system has certain limitations.

The effectiveness of AI-driven matching depends heavily on the completeness and accuracy of user profile data. If students or alumni provide limited or outdated information, recommendation precision may decrease. Additionally, in highly specialized domains, human oversight may still be

required to ensure optimal mentor allocation. Despite these limitations, the system demonstrates strong potential for adoption in modern higher education institutions. Its scalable architecture and centralized monitoring capabilities make it suitable for institutions seeking structured alumni engagement.

The platform transforms alumni networking from a passive record-keeping system into an active, intelligent mentorship ecosystem. The evaluation results emphasize the advantages of automation in alumni management processes. Traditional systems rely heavily on manual coordination, which can lead to delays and mismatches. The automated matching system ensures uniform evaluation criteria for all users, reducing bias and improving fairness in mentor allocation. The ability to generate ranked mentor recommendations significantly enhances efficiency by providing personalized suggestions based on measurable compatibility factors. Although effective, the system should complement institutional oversight rather than completely replace human judgment, ensuring a balanced and efficient alumni engagement model.

IX. CONCLUSION AND FUTURE WORK

The College Alumni Connect and Management System demonstrates how artificial intelligence can transform alumni relationship management into a structured and intelligent process. By integrating similarity-based skill matching with centralized database management, the system improves mentorship accuracy and engagement efficiency. The platform enhances communication between students and alumni while ensuring secure and organized data handling. The system enables institutions to leverage alumni expertise more effectively, resulting in improved career guidance and professional exposure for students. Automated mentor recommendation reduces administrative workload and minimizes mismatches while centralized dashboards provide measurable insights into participation trends and mentorship outcomes.

Over time, the AI module refines its recommendations based on updated data and feedback, thereby improving overall system performance. Future development of the system may include integration with mobile applications to increase accessibility and user convenience. The implementation of real-time messaging features could further enhance interaction between mentors and mentees. Advanced predictive analytics may be incorporated to forecast career trends and suggest mentorship paths accordingly.

The system can also be expanded to support industry partnerships and internship recommendations based on skill compatibility. Incorporating advanced machine learning models could further enhance recommendation accuracy in complex specialization domains. Overall, the proposed system provides a scalable, secure, and intelligent solution for modern alumni management with strong potential to improve student career outcomes and strengthen long-term institutional relationships.

X REFERENCES

- [1] R. S. Pressman, *Software Engineering: A Practitioner's Approach*, 8th ed. New York, NY, USA: McGraw-Hill, 2014.
- [2] K. G. Jagtap et al., "Alumni Connect: Bridging Past and Present," *International Journal of Research Publication and Reviews (IJRPR)*, 2024.
- [3] L. L. Occiano et al., "Web-Based Alumni Management

System," *International Journal of Research in Engineering and Science (IJRES)*, 2023.

[4] "Digital Alumni Platforms in Universities," *The Hindu*, 2023.

[5] *Software Engineering Practices for Web Systems*, Technical Report, 2022.